

ArtData™ Module

AD-P — ArtData™ Provenance Module

Module ID: AD-P

Standard System: ArtData™

Category: AI & Data Integrity Standards (AI)

Subcategory: Dataset Provenance

Version: 1.0

Status: Canonical · Module

Compatibility: ArtData™ Standard · MTVF™ · AI Governance Frameworks

About the Module

What the Provenance Module Is

The **ArtData™ Provenance Module (AD-P)** defines the structural conditions required to document the origin of datasets used in artificial intelligence systems.

AI models depend heavily on training data, yet in many environments dataset origin remains unclear or partially documented.

The Provenance Module establishes minimum transparency conditions that allow dataset sources to be identified, traced, and verified.

It ensures that datasets used in AI systems have **visible and accountable origin structures**.

What This Module Changes

In many AI environments datasets are:

- aggregated from multiple sources
- partially undocumented
- difficult to trace back to their origin

The Provenance Module introduces a structural principle:

every dataset must have a documented origin structure.

This transforms datasets from opaque resources into **traceable data assets**.

Canonical Definition

ArtData™ Provenance is the structural documentation of dataset origin, including the source category, acquisition method, and initial creation context associated with a dataset used in AI systems.

Dataset provenance establishes the traceable starting point of a dataset lifecycle.

Scope

The Provenance Module applies to datasets used in:

- AI training systems
- machine learning evaluation datasets
- simulation environments
- AI testing frameworks
- automated decision system training pipelines

The module focuses exclusively on **dataset origin structures**.

Structural Requirements

To satisfy ArtData™ Provenance conditions, the following information must be documented.

Source Identification

The dataset source must be identifiable.

Examples include:

- internally generated dataset
 - licensed commercial dataset
 - publicly available dataset
 - sensor-generated dataset
 - research dataset.
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Acquisition Method

The method through which the dataset was obtained must be declared.

Examples:

- direct collection
 - licensing agreement
 - research collaboration
 - automated data capture.
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Dataset Origin Record

The initial dataset creation or acquisition event must be recorded.

Minimum information:

- acquisition date
 - acquisition method
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- source reference.
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Minimum Implementation Framework (MIF)

Implementation Steps

Step 1 — Identify Dataset Source

The dataset source category must be declared.

Example categories:

- internal data generation
 - public dataset
 - licensed dataset
 - sensor-generated data.
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Step 2 — Record Acquisition Method

Document how the dataset entered the system.

Examples:

- purchased dataset
 - downloaded public dataset
 - internal data collection.
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Step 3 — Create Provenance Record

Create a dataset origin record containing:

- dataset identifier
 - source description
 - acquisition method
 - acquisition date.
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Step 4 — Store Provenance Metadata

The provenance information must be stored in:

- dataset metadata
 - or
 - dataset registry
 - or
 - data governance documentation.
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Architecture Position

The Provenance Module represents the **first layer of the ArtData™ architecture.**

ARTDATA™ DATA INTEGRITY STRUCTURE

Dataset Origin (AD-P)



Dataset Identity



Lifecycle Continuity



Transformation Transparency



Responsible Entity

Dataset provenance defines the **starting point of dataset accountability**.

Use Case 1

AI Training Dataset Source Transparency

An AI startup trains models using multiple datasets.

By applying the Provenance Module, the organization documents:

- dataset origin
- acquisition method
- dataset creation event.

Result:

- training data becomes traceable
 - internal governance improves
 - dataset transparency increases.
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Use Case 2

Research Dataset Reproducibility

A research institution publishes AI models trained on multiple datasets.

Without provenance documentation, it may be impossible to reproduce the experiment.

By implementing the Provenance Module:

- dataset sources are documented
- acquisition methods are recorded
- dataset origin becomes verifiable.

Result:

- reproducible research environments
 - stronger academic credibility.
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Canonical Closing Statement

Reliable AI begins with traceable data origins.

The ArtData™ Provenance Module establishes the starting point of

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Canonical Source

<https://originopenfoundation.org/>